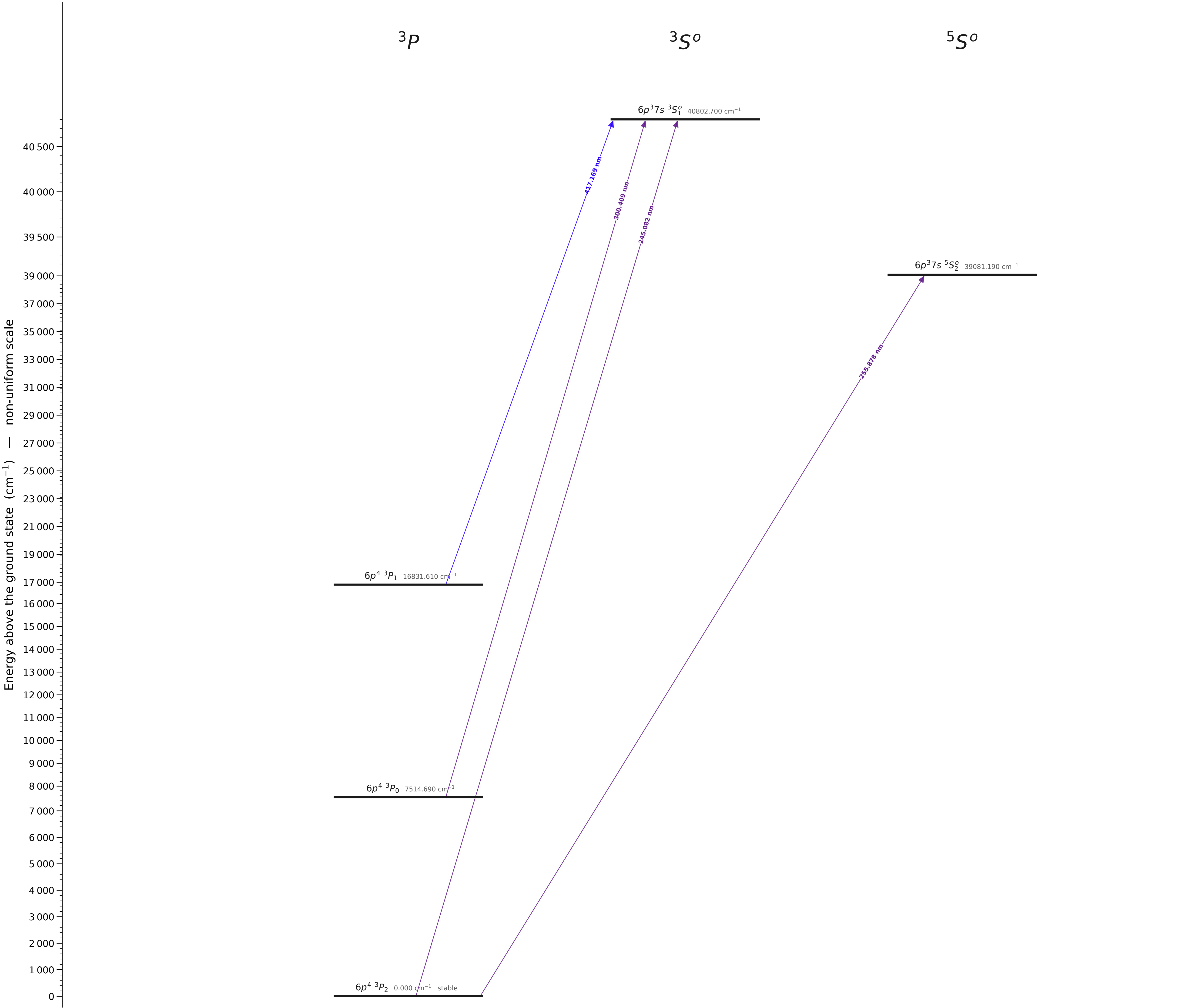


Polonium

Po (Po I). Strongest classified transitions below 2 μm (NIST): 5 levels, 4 lines.

Po⁺ ionisation limit 67896.31 cm⁻¹ = 8.41807 eV (147.283 nm)



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
$6p^4\ ^3P_2 - 6p^3 7s\ ^5S_2^o$	255.8776	255.8008	1171.6246	-	-	-	-	first step of the ISOLDE-RILIS polonium resonance ionizat...

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

²⁰⁷Po hyperfine structure (*I* = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)				
$6p^4\ ^3P_2$	139.5(3)	-	F=1/2: -976.50	F=3/2: -767.25	F=5/2: -418.50	F=7/2: +69.75	F=9/2: +697.50
$6p^3 7s\ ^5S_2^o$	564.0(10)	367(7)	F=1/2: -3691.10	F=3/2: -3010.25	F=5/2: -1783.75	F=7/2: +126.03	F=9/2: +2911.75

Measured A, B constants; sources in the data file.

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where available).